

11 RADIATION AND CONTACT HEAT TRANSFER

Most of the MARS variables for the radiation heat transfer model are stored in the "rstplt" file. Thus, a restart calculation with the same radiation heat transfer input is possible using the "rstplt" file of the previous run. A complete change of the radiation heat transfer input is also allowed in "restart" calculations. However, "partica"l changes of the radiation heat transfer input are not allowed in "restart" calculations.

11.1 CARD 70000000, Radiation and Contact Heat Transfer Model Control Card

Any heat structure may radiate or contact to any other heat structure or set of heat structures in a user-defined enclosure. An enclosure is a set of heat structures that communicate via thermal radiation and contact. The calculation ignores fluid in the enclosure.

W1(I) Number of sets of radiation/contact enclosures, nset. This word must be less than 100 and greater than 0 for radiation to be on.

11.1.1 Card 7SS00000, Radiation/Contact Set Card

SS is the set number. One of these cards must be input for each radiation/contact set. The first word is required; the other three are optional.

W1(I) Number of radiating/contact heat slabs, nrh. This word is the number of heat slabs (surfaces) that participate in radiation/contact heat transfer in set SS. The maximum is 99. If this number is positive, Format 1 is used for Cards 7SSNN101 through 7SSNN199 and no contact heat transfer condition is adopted. If this number is negative, Format 2 is used for Cards 7SSNN101 through 7SSNN199.

W2(R) Minimum temperature, trmin (K, °F). This word is the minimum temperature of all surfaces in a radiation enclosure below which radiation will no longer be calculated. The default value is 900 K.

W3(R) Minimum void fraction, voidmn. This word is the minimum void fraction below which radiation will no longer be calculated. The default value is 0.75.

W4(I) View factor set, refset. If input, this is the number of the set (SS) from which view factors are to be obtained. If no number is found here, the view factors must be input for this set.

11.2 Cards 7SSNN001 through 7SSNN099, Radiation/Contact Heat Structure Data

For these cards, SS must take on every value from 1 to nset (Word 1 in Card 50000000), and NN must take on every value from 1 to nrh (Word 1 on Card 5SS00000) for each SS. Data are entered for each conductor surface that participates in radiation/contact heat transfer.

- W1(I) Heat structure geometry level, jrh. This word is CCCG0ZZ, where CCCG is the heat structure geometry combination of nh, and ZZ is the axial level number participating in radiation/contact.
- W2(I) Surface flag, jlr. For this word, 0 = left surface, and 1 = right surface of NN.
- W3(R) Emissivity of surface NN.

11.2.1 Cards 7SSNN101 through 7SSNN199, Radiation View Factors and Contact Heat Transfer Coefficients

There are nrh•nrh values in each set. SS is the set number (from 1 to nset). NN is the surface number (from 1 to nrh). For a given NN, the sum of the view factors must be 1.0, and the view factor times the surface area must equal the view factor times the area of the receiving surface.

Format 1: When W1 of Card 7SS00000 is positive

- W1(R) View factor, vfij. View factor from surface NN to surface W2(I).
- W2(I) Radiation/contact surface number to which NN radiates or contacts. Repeat the above two words until view factors to all nhr surfaces from all surfaces are entered. Sequential expansion is used.

Format 2: When W1 of Card 7SS00000 is negative

- W1(R) View factor, vfij. View factor from surface NN to surface W3(I).
- W2(R) Effective contact heat transfer coefficient between surface NN and surface W3(I) (W/m².K).
- W3(I) Radiation/contact surface number to which NN radiates or contacts. Repeat the above two words until view factors to all nhr surfaces from all surfaces are entered. Sequential expansion is used.